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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

- Let It Be AI is a web-based artificial intelligence platform designed to perform advanced image editing and enhancement operations using modern AI technologies.
- The system provides multiple image transformation features such as Image Restoration, Background Removal, Object Removal, Object Recoloring, and Generative Fill. These functionalities help users edit and improve images without requiring professional graphic design skills.
- The project is developed using Next.js and TypeScript for the frontend and backend framework, MongoDB for database management, Cloudinary for image storage and AI-based transformations, and NextAuth for secure user authentication.
- The platform follows a credit-based model where users can purchase credits and utilize them for image transformations. Razorpay integration enables secure online payments.
- The primary objective of the project is to make AI-powered image editing accessible, efficient, and user-friendly while demonstrating the practical application of artificial intelligence in multimedia processing.

1.2 LITERATURE REVIEW

- Recent advancements in Artificial Intelligence and Computer Vision have significantly improved image editing and enhancement technologies.
- Cloud-based AI services provide automated image processing capabilities such as object detection, background segmentation, image restoration, and generative content creation.
- Modern web frameworks such as Next.js enable developers to create scalable and responsive applications with efficient server-side rendering and routing.
- Previous research and commercial solutions such as Adobe Firefly, Canva AI, and Cloudinary AI transformations demonstrate the growing demand for intelligent image editing platforms.
- Secure authentication systems and cloud databases have further enabled the development of personalized and scalable AI applications for a large number of users.

1.3 PROBLEM STATEMENT

Despite the availability of image editing software, many existing solutions:

- Require advanced technical knowledge and design expertise.
- Consume significant system resources and installation space.
- Offer limited automation for image enhancement tasks.
- Have expensive subscription models for premium AI features.

The objective of this project is to:

- Develop a web-based AI image editing platform.
- Provide automated image transformation features through an easy-to-use interface.
- Enable secure user management and credit-based service utilization.
- Deliver scalable and cloud-based image processing solutions.

CHAPTER 2

METHODOLOGY

The development of Let It Be AI follows the following methodology:

- **User Authentication:**

Users register and log in securely using NextAuth authentication mechanisms.

- **Image Upload:**

Users upload images through the web interface. Images are stored and managed using Cloudinary cloud storage.

- **AI Image Processing:**

The selected transformation is applied using Cloudinary AI services and image processing APIs.

→ **Image Restoration:** Enhances damaged or low-quality images.

→ **Background Removal:** Automatically removes image backgrounds.

→ **Object Removal:** Detects and removes unwanted objects from images.

→ **Object Recoloring:** Changes the color of selected objects.

→ **Generative Fill:** Expands or modifies image regions using AI-generated content.

- **Credit Management:**

Each transformation consumes credits from the user's account. Credit balances are maintained in MongoDB.

- **Payment Integration:**

Razorpay is integrated to allow users to purchase additional credits securely.

- **Database Management:**

User profiles, transaction records, image details, and credit information are stored in MongoDB.

- **Administrative Control:**

An admin panel provides monitoring and management of users and transactions.

CHAPTER 3

RESULTS AND DISCUSSION

The developed system was tested successfully in a web environment and produced the following results:

- Successfully authenticated users and managed account information.
- Uploaded and processed images efficiently through cloud services.
- Performed AI-powered image transformations with high accuracy.
- Maintained transaction records and user credit balances correctly.
- Provided responsive and user-friendly interfaces across different devices.

Limitations:

- ☆ Image processing speed depends on internet connectivity and cloud service response time.
- ☆ Advanced transformations may consume higher computational resources.
- ☆ Dependence on third-party AI services may affect performance during service interruptions.

CHAPTER 4

CONCLUSION

4.1 CONCLUSION

- Let It Be AI successfully demonstrates the integration of Artificial Intelligence, Cloud Computing, and Modern Web Technologies into a practical image editing platform.
- The system provides intelligent image transformation capabilities through an easy-to-use web interface, reducing the complexity of traditional image editing workflows.
- The project highlights the effectiveness of AI-powered image processing in improving productivity and accessibility for users.
- Overall, the platform serves as a scalable, secure, and efficient solution for AI-based image enhancement and transformation tasks.

4.2 FUTURE SCOPE

- Integration of advanced generative AI models for image generation and editing.
- Support for video enhancement and transformation features.
- Mobile application development for Android and iOS platforms.
- Real-time collaborative editing and sharing functionality.
- AI-powered image captioning and content analysis.
- Implementation of additional payment gateways and subscription plans.
- Enhancement of personalization through machine learning-based recommendations.

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